DAY 18 | POST-READ

Context and RAG: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Context engineering	Decide what goes in: instructions, input, history, knowledge, tools, examples.
Principles	Relevant over complete, budget tokens, order matters, label sources, compress history.
Context rot and compaction (NEW v4)	Long, noisy context hurts quality; summarise old turns, keep stated facts.
AGENTS.md and Skills (NEW v4)	Persistent context; Skills load full instructions only when needed.
Reference architecture (NEW v4)	AI gateway, MCP tools, identity, retrieval, evals, tracing.
RAG pipelines	Ingestion: load, clean, chunk, embed, store. Query: retrieve, assemble, generate, cite.
Permission-aware retrieval (NEW v4)	Filter chunks by the user's rights before ranking.
Injection via documents (NEW v4)	Retrieved text is data; scan at ingestion, wrap and restate rules at query.
Grounded prompt	Use only the context; say "I don't know"; cite chunk ids; low temperature.
Chunk size	Start around 300 to 800 tokens, 10 to 20 percent overlap, then measure.

Part B: Going beyond today

1. Lost in the middle

Studies of long-context models have found that information placed in the middle of a long context is used less reliably than information near the start or end. Put the most relevant chunks first, and keep context lean.

2. Parent-child chunking

Index small chunks for precise matching, but send the larger parent section to the model so it has full context. Many RAG frameworks support this pattern.

3. Contextual chunk headers

Prefixing each chunk with its document and section titles helps both retrieval and the model's understanding.

```
chunk_text = f"Document: Telecom FAQ > Section: Roaming\n{chunk}"
```

4. Cache-friendly context

Put stable parts (rules, tool list, AGENTS.md content) first and changing parts (retrieved chunks, the question) last. Providers can then reuse the cached prefix and charge less. More on Day 25.

5. When not to use RAG

If the knowledge is small and stable, put it straight into the prompt. If the task is about style or format rather than facts, examples may suit better. RAG shines for large, changing knowledge. Day 22 compares RAG, fine-tuning, long context and caching.

Part C: Check yourself

- Name five things that can go into a context window.
- What is context rot, and how does compaction help?
- What belongs in AGENTS.md, and what must never go in it?
- Why must permission filtering happen before retrieval ranking?
- How can a document attack an AI assistant, and name two defences.

Part D: Practice ideas for tonight

- Chunk the telecom FAQ three ways and compare the retrieved chunks for the same question.
- Write a SKILL.md for one SmartDesk task and keep its description under 25 words.
- Add one poisoned FAQ entry and check that your scanner quarantines it.

Coming up on Day 20: Git, CI/CD with security gates, and Docker.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today